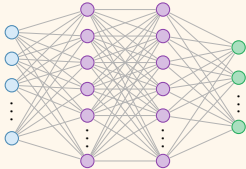


Actor

Cost Network π_θ



$\mathbf{x}$

θ_{MPC}

MPC Solver π_{MPC}

$$\begin{aligned} & \arg \min_{\mathbf{u}} \sum_{k=0}^{N-1} L_k(\mathbf{x}_k, \mathbf{u}_k; \theta_{\text{MPC}}) + M(\mathbf{x}_N; \theta_{\text{MPC}}) \\ & \text{s.t. } \mathbf{x}_0 = \mathbf{x} \\ & \quad \mathbf{x}_{k+1} = f_k(\mathbf{x}_k, \mathbf{u}_k; \theta_{\text{MPC}}) \\ & \quad 0 \geq h_k(\mathbf{x}_k, \mathbf{u}_k; \theta_{\text{MPC}}) \\ & \quad 0 \geq h_N(\mathbf{x}_N; \theta_{\text{MPC}}) \end{aligned}$$

$\mathbf{u}$